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MAST20030: Differential Equations
Revision Tutorial Solutions: Week 1

Solution to Question 1.

(a) Yes. We can see this by constructing the linear system $\alpha_1 \begin{bmatrix}1\\1\end{bmatrix} + \alpha_2 \begin{bmatrix}2\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$. We immediately have that $\alpha_1 = 0$ is the only solution for α_1 and hence $\alpha_2 = 0$ is the only solution for α_2.

(b) $\mathbb{R}^2 = \left\{ \begin{bmatrix}\alpha_1\\\alpha_2\end{bmatrix} \mid \alpha_1, \alpha_2 \in \mathbb{R} \right\}$. Now $\text{span}\left(\left\{ \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}2\\0\end{bmatrix} \right\} \right) = \left\{ \begin{bmatrix}\beta_1 + 2\beta_2\\\beta_1\end{bmatrix} \mid \beta_1, \beta_2 \in \mathbb{R} \right\}$. By letting $\alpha_1 = \beta_1 + 2\beta_2$ and $\alpha_2 = \beta_1$ we can see that these two sets are equivalent and hence $\mathbb{R}^2 = \text{span}\left(\left\{ \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}2\\0\end{bmatrix} \right\} \right)$. Because these two vectors are linearly independent, and they span $\mathbb{R}^2$, they meet the definition of a basis and hence the dimension of $\mathbb{R}^2$ is 2.

Solution to Question 2.

(a) $\text{span}(\{f, g, h\}) = \{\alpha_1 \sin x + \alpha_2 \cos x + 2\alpha_3 \sin x \mid \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}\}$

(b) $h = 2f$ and hence we have a solution to $\alpha_1 f + \alpha_2 g + \alpha_3 h = 0$ in which not all of $\alpha_1, \alpha_2, \alpha_3$ are zero, e.g., $\alpha_1 = 2, \alpha_2 = 0, \alpha_3 = -1$.

(c) There is no non-zero solution to $\alpha_1 \sin x + \alpha_2 \cos x = 0$ and hence f and g are linearly independent.

(d) Because h is a multiple of f, we have that $\text{span}(\{f, g, h\}) = \text{span}(\{f, g\})$. When combined with the fact that f and g are linearly independent we have that $\{f, g\}$ is a basis for $\text{span}(\{f, g, h\})$, which will therefore have dimension 2.

Solution to Question 3.

Substituting $e^{\lambda t}$ into the ODE gives

$$(\lambda^2 + 3\lambda + 2)e^{\lambda t} = 0$$

and so $\lambda = -1, -2$. The general solution to the ODE therefore is $x(t) = C_0 e^{-2t} + C_1 e^{-t}$. Applying the initial conditions gives $C_0 = -x_0 - v_0$ and $C_1 = 2x_0 + v_0$ and so the solution to the initial value problem is

$$x(t) = -(x_0 + v_0)e^{-2t} + (2x_0 + v_0)e^{-t}$$

Solution to Question 4.

(a) $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \lambda^2 + 3\lambda + 2 = 0$ and therefore the eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -2$. For the first eigenvector: $\begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \boldsymbol{v}_1 = \boldsymbol{0}$ and so $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector for λ_1. For the second eigenvector: $\begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \boldsymbol{v}_2 = \boldsymbol{0}$ and so $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is an eigenvector for λ_2.

(b) Yes the two vectors are linearly independent (check that the only solution to $\alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 = \boldsymbol{0}$ is $\alpha_1 = \alpha_2 = 0$).

(c) $\mathbb{R}^2 = \text{span}\,(\{\boldsymbol{v}_1, \boldsymbol{v}_2\})$ (see solution from 1(a) for the procedure to show this). Hence $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ is a basis for $\mathbb{R}^2$.

Solution to Question 5.

(a) Let $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ therefore implies that $x_1 - x_2 = 0$ and $-x_1 + x_2 = 0$. Both of these equations are just $x_1 = x_2$ so any vector of the form $\boldsymbol{x} = \begin{bmatrix} t \\ t \end{bmatrix}$ where $t \in \mathbb{R}$ will be a solution. The solution space is therefore given by span $\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$ and hence the dimension of the solution space is 1.

(b) Given that $\boldsymbol{b}$ is just the first column of $\boldsymbol{A}$, we have that $\boldsymbol{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boldsymbol{b}$, i.e., $\boldsymbol{y}_p = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

(c) The general solution to $\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b}$ is therefore $\boldsymbol{y} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ where $t \in \mathbb{R}$.

Solution to Question 6.

(a) $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \lambda^2 - 4\lambda + 3 = 0$ and so the two eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$. For the first eigenvector: $\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \boldsymbol{v}_1 = \boldsymbol{0}$ and so $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector for λ_1. For the second eigenvector: $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \boldsymbol{v}_2 = \boldsymbol{0}$ and so $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector for λ_2.

(b) $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = 1 \times 1 + (-1) \times 1 = 0$ and so $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are orthogonal.

(c) Let $\boldsymbol{w} \in \mathbb{R}^2$. Because $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ is a basis for $\mathbb{R}^2$, we can write $\boldsymbol{w} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2$. Now taking the dot product of both sides with $\boldsymbol{v}_1$ gives $\boldsymbol{w} \cdot \boldsymbol{v}_1 = \alpha_1 \boldsymbol{v}_1 \cdot \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 \cdot \boldsymbol{v}_1$ and so $\alpha_1 = \dfrac{\boldsymbol{w} \cdot \boldsymbol{v}_1}{\boldsymbol{v}_1 \cdot \boldsymbol{v}_1} = \dfrac{1}{2} \boldsymbol{w} \cdot \boldsymbol{v}_1$. Similarly we can show that $\alpha_2 = \dfrac{1}{2} \boldsymbol{w} \cdot \boldsymbol{v}_2$.